Metadata for Knayton (KNA), Thirsk, UK

Description of data set	Magnetic time-series from the BGS SWIGS differential magnetometer method (DMM) systems. Funded by NERC, grant number: NE/P017231/1 "Space Weather Impact on Ground-based Systems (SWIGS)". These data consist of measurements of the Earth's natural magnetic field at the remote site (KNAR) and the natural magnetic field plus the field created by GIC at the underline site (KNAU). The database will include .xyz files with the DMM data and one document with metadata. See Hübert et al. (2020) for further details.
Location (in lat/long): underline station	54.302096 N, -1.314834 E
Location (in lat/long): remote station	54.302347 N, -1.319507 E
Distance between DMM sites	305 m
Description of site (i.e. open moorland)	Fallow field between enclosed grazing lands.
Equipment used	Sensys 3DFM fluxgate magnetometer and Earth Data digitisers (EDR 209, 24bit)
Installation date	31/10/2018
Removal date	02/09/2019
Number of days collected	307
Data sampling rate	1 Hz
Notes of known data gaps	KNAR: Data gap due to cable damage: 11/03/2019-05/06/2019. Data stops at KNAR on 22/08/2019.
Qualitative estimate of site quality	Very good.
Orientation angle of HV wires relative to true north	-10.2° (i.e. lines run approximately N-S)
Alignment of magnetometers	Magnetic north as of 09/08/2018
Voltage level of the lines (Thornton – Lackenby)	400 kV
Height above the ground of each wire (m)	Lower: 16.39 m, Middle: 25.33 m, Upper: 36.38 m
Underline magnetometer	Central (9m to either circuit)
File Type	ASCII text (.xyz)
File format	YYYY-MM-DD HH:MM:SS X Y Z Five columns with 86400 rows (plus one header row). Magnetic values in units of nT (10^-9 T). Empty values denoted by: 999999.00
Citation	Hübert, J., Beggan, C. D., Richardson, G. S., Martyn, T., & Thomson, A. W. P. (2020). Differential magnetometer measurements of geomagnetically induced currents in a complex high voltage network. Space Weather, 18, e2019SW002421. doi: 10.1029/2019SW002421